

HAT-P-32b

R-0272

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_250929 · created 2026-07-17T13:41:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460947.8708 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.0091
Transit depth (Rp/Rs)^2	2.5 +/- 0.29 [%]
Orbital Inclination (inc)	87.51 +/- 1.56
Airmass coefficient 1 (a1)	1.1644 +/- 0.0054
Airmass coefficient 2 (a2)	-0.136 +/- 0.036
Scatter in the residuals of the lightcurve fit is	0.46 %
Best Comparison Star	None
Optimal Aperture	3.87
Optimal Annulus	8.77
Transit Duration (day)	0.1031 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.0091 vs 0.1489 (deviation 1.0 σ)
Duration fit vs published	0.1031 d vs 0.1304 d (deviation 10.93 σ)
Mid-time O-C	-1.4 min
Depth SNR	17.4
Residual scatter	0.46 %

Light curve

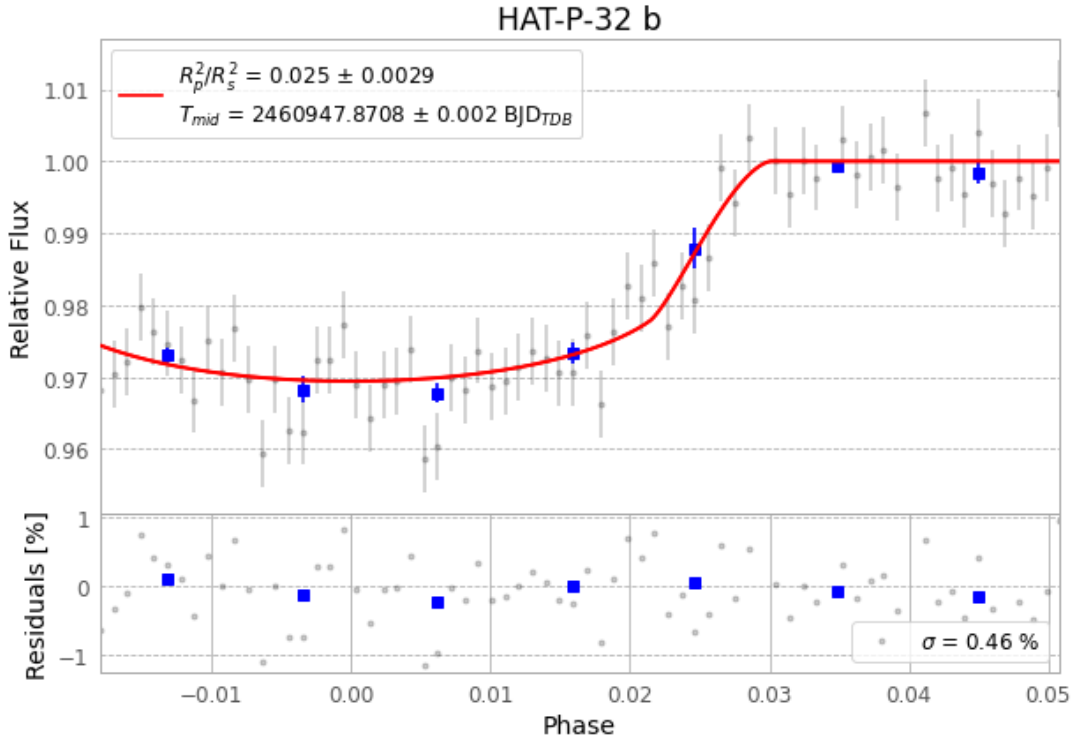


Figure 1 — FinalLightCurve_HAT-P-32 b_2025-09-29.png (SHA-256 9ac25fe43fd51dda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 323], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 42b28669bae45923...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 5cfc163

Data provenance

72 FITS frames (47 MB), HATP-32250929075115.FITS → HATP-32250929112415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-250929.log (77df46c685c0...), FinalLightCurve_HAT-P-32 b_2025-09-29.png (9ac25fe43fd5...), FinalLightCurve_HAT-P-32 b_2025-09-29.csv (2885ce625550...)

Compute resources

Wall time 90.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 13:41Z.